

TCP HTCP Metrics Analysis

HTCP (Hamilton TCP) is a congestion control algorithm designed to improve TCP performance in high-bandwidth, long-latency networks. It aims to optimize throughput and fairness, especially in environments with large bandwidth-delay products (high-speed, long-distance connections).

Key Concepts of HTCP

1. Adaptive Congestion Control:

- HTCP adjusts its congestion window growth rate based on the **elapsed time since the last congestion event**.
- This allows HTCP to be more aggressive when there is no congestion and less aggressive when congestion occurs.

2. Alpha and Beta Parameters:

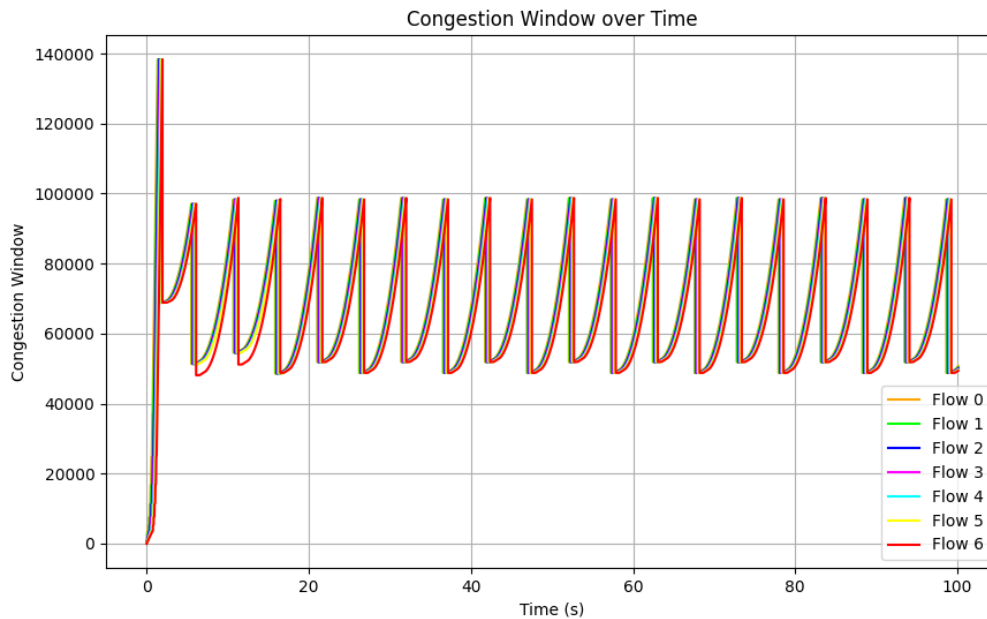
- **Alpha (α)**: Controls the rate of congestion window increase in congestion avoidance.
 - If there is no congestion for a while, α increases to grow the CWND faster.
- **Beta (β)**: Controls the reduction of the congestion window during congestion.
 - HTCP adjusts β dynamically based on throughput changes and RTT.

3. Throughput and RTT Awareness:

- HTCP tracks metrics like **minimum RTT**, **maximum RTT**, and **throughput** to adaptively tune α and β .

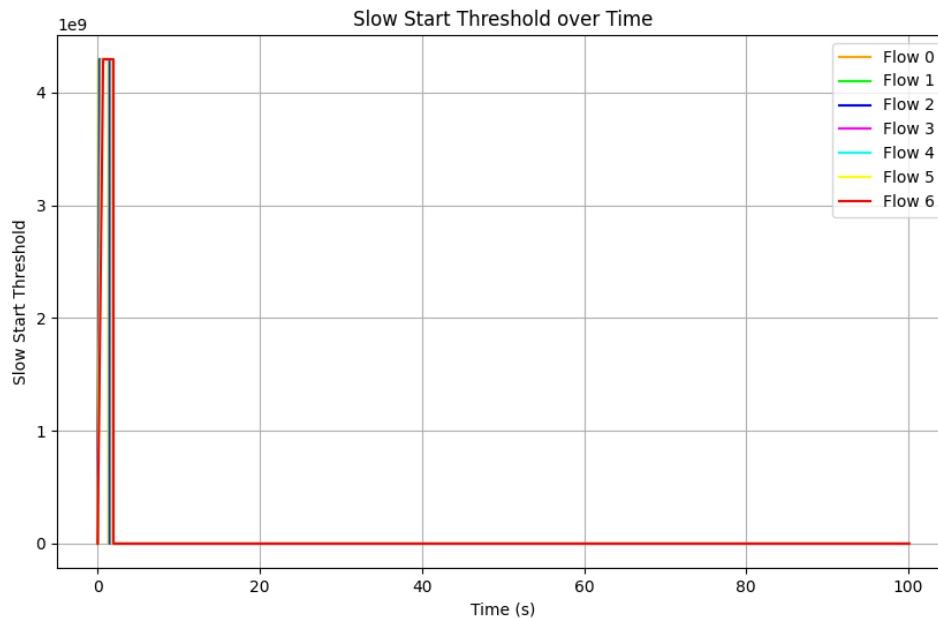
Graphs

Congestion Window (cwnd): A TCP state variable that controls the amount of data a TCP sender can send into the network before receiving an acknowledgment (ACK) for previously sent data.



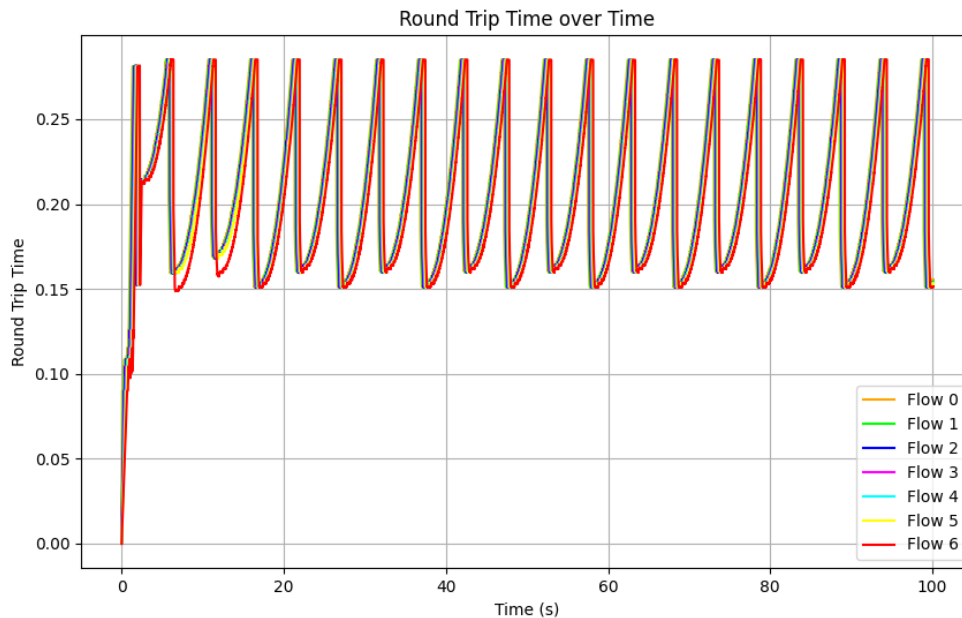
- The CWND grows **exponentially** during the initial phase (slow start).
- After congestion is detected, CWND decreases based on m_beta and recovers **linearly** in the congestion avoidance phase.
- The growth of CWND depends primarily on m_alpha

Slow Start Threshold(ssthresh): The slow start threshold determines when TCP switches from aggressive exponential growth (slow start phase) to more cautious linear growth (congestion avoidance phase). It's like a speed limit that TCP sets for itself based on previous congestion experiences. When cwnd reaches ssthresh, TCP becomes more conservative in its growth.



- The Slow Start Threshold (SSTH) starts at an extremely high value, reflecting the **initial aggressive behavior** of the Hamilton TCP (H-TCP) variant.
- Once the congestion window (cwnd) exceeds the Slow Start Threshold (SSTH), the TCP variant no longer requires SSTH for further growth.
- SSTH remains at zero because it is no longer used or updated in congestion avoidance mode, where window growth is driven by the **updateAlpha function** and **adaptive mechanisms**.

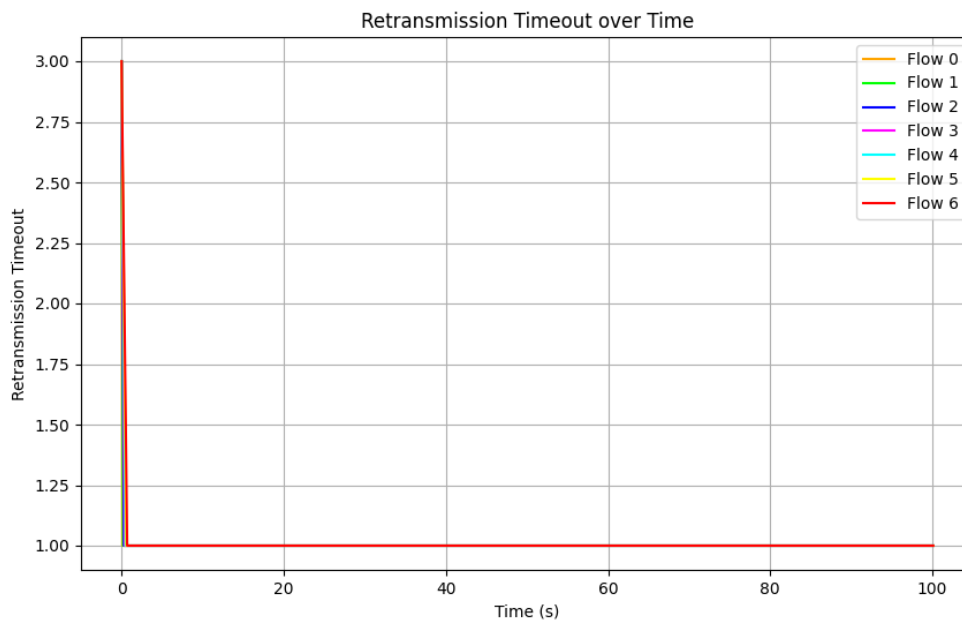
Round Trip Time (RTT): RTT measures how long it takes for a packet to reach its destination and for an acknowledgment to return.



- At the start of the connection, RTT is relatively stable or slightly variable as the network adapts to the initial burst of traffic during the **slow start phase**.
- When the network becomes congested, RTT often spikes due to increased queuing delays in intermediate routers or switches.
- After congestion is detected and adjustments (e.g., **window reduction**) are made, RTT stabilizes as the network operates in a more balanced state.

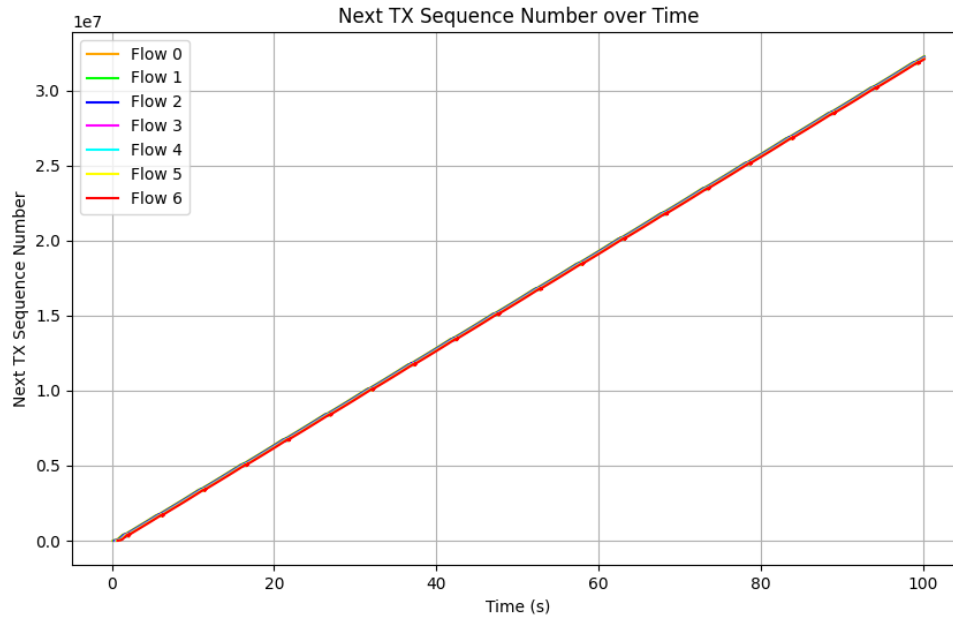
alpha increases → CWND growth → queue buildup → RTT increase → beta increases → CWND reduction → queue drain → RTT decrease

Retransmission Timeout (RTO): RTO determines how long TCP waits before assuming a packet is lost and needs retransmission. It's like a safety timer - if no acknowledgment arrives within this time, TCP assumes the worst and retransmits. The RTO value is typically calculated based on RTT measurements and their variance.



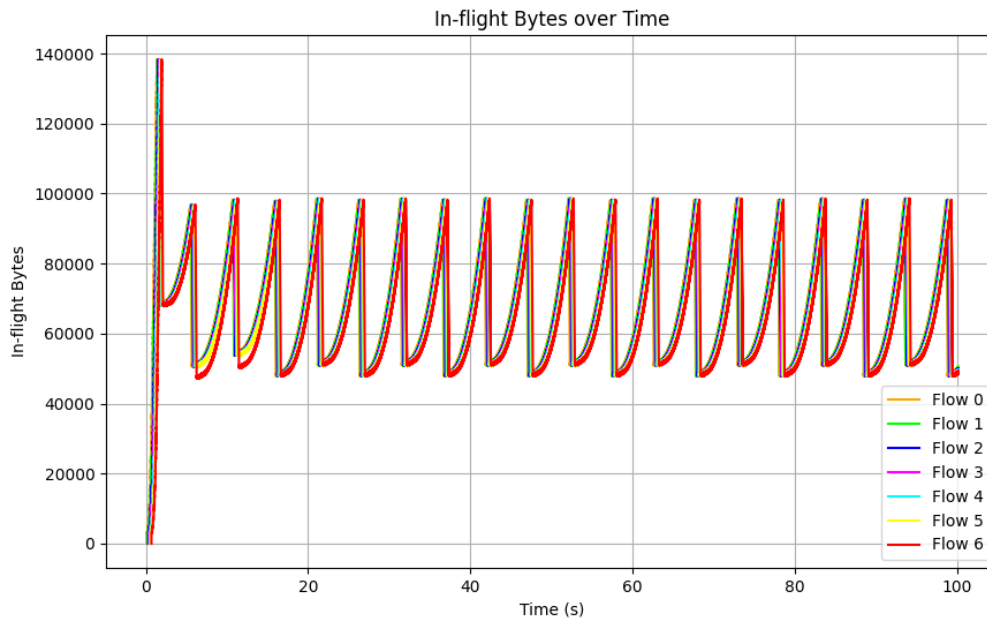
- At the start (near 0 seconds), there is a sharp spike in the RTO, reaching **3.0**. This typically indicates initial packet loss or high RTT variance, causing a conservative timeout estimate.
- The RTO quickly drops to a stable value of **1.0**. This signifies that network conditions stabilize after the initial adjustment
- For the remainder of the simulation (0–100 seconds), the RTO remains constant at **1.0**, showing no additional packet loss, retransmission events, or significant RTT variations

Next Transmission Sequence Number (next-tx): This tracks the sequence number of the next packet to be transmitted. It's like a counter that helps ensure packets are sent in order.



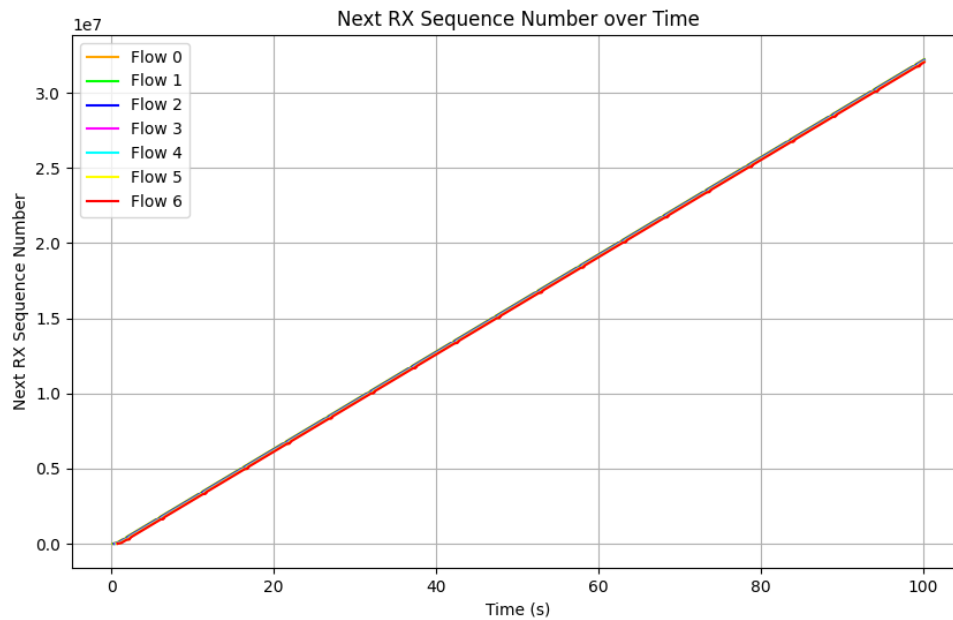
- Smooth growth, based congestion window size.
- Gentle backoff during congestion.
- Better network stability under varying RTT and congestion scenarios.

In-flight Bytes: This measures how many bytes are currently in transit - sent but not yet acknowledged. It's like tracking packages that are currently in delivery.



- At the start (near 0 seconds), there is a sharp increase in in-flight bytes due to the **slow-start phase**, where the congestion window (cwnd) grows exponentially.
- The graph exhibits a repetitive sawtooth pattern, typical of **congestion avoidance** in TCP.
- The sudden dips in in-flight bytes represent **packet losses or RTT-triggered congestion events**, where the cwnd resets and the recovery process begins.

Next Receive Sequence Number (next RX): This indicates which sequence number the receiver expects next. It helps detect missing or out-of-order packets.



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- Gentle backoff during congestion.
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